

EE 534: Probability and Random Processes

Course Code	EE 534
Course Name	Probability and Random Processes
L-T-P-C	3-0-0-3 (L-T-P-C)
Prerequisites	Linear algebra (MA512 or IC 111) and IC 252 or equivalent of both (MA512 and IC252)
Intended for	UG /MS/MTech (CSP)/PhD/M.Sc(Maths)
Distribution	Core for Mtech (CSP), Elective for B.Tech. III/IV year, MS, Ph.D, M.Sc(Maths). Semester : Odd
Approval	22nd Senate (Jun 2019)
Recorded from	CS 601 by the 22nd Senate (Jun 2019)

Course Contents

- **Module I:** Sigma field. Review of - axiomatic probability, conditional probability and independence. [2 Lectures]
- **Module II:** Recap of random variables and functions of random variables. [3 Lectures]
- **Module III:** Probability generating function, moment generating function and characteristic functions properties and applications. [3 Lectures]
- **Module VI:** Markov chains, classification of states and chains, stationary distribution and limit theorem, Poisson process. [5 Lectures]
- **Module V:** Convergence of random variables basic results, inequalities (Markov and Chebyshev), law of large numbers (weak and strong), central limit theorem. [5 Lectures]
- **Module VI:** Concentration inequalities Chernoffs bound, Hoeffdings inequality, Bennetts inequality, Bernstiens inequality and Efron-Stein inequality. [8 Lectures]
- **Module VII:** Random vectors and covariance matrix. Random processes stationarity, WSS. Autocorrelation, cross correlation, power spectral density. Filtering of WSS processes. Basic notion of ergodicity. Wiener processes, Markov processes. [10 Lectures]
- **Module VIII:** Queueing models - Little's law, M/M/1, M/M/m, M/M/m/m, M/G/1 queueing systems, priority queueing. [6 Lectures]

Text Books

1. Grimmett and Stirzaker, **Probability and Random Processes**, Oxford University Press, 2001.

References

1. Erhan Cinlar, **Introduction to Stochastic Processes**, Dover Books on Mathematics, 2013
2. R. G. Gallager, **Stochastic Processes: Theory for applications**, Cambridge University Press, 2014.
3. S. M. Ross, **Stochastic processes**, 2nd Edition, John Wiley, 1996.
4. J. R. Norris, **Markov chains**, Cambridge University Press, 1999.
5. Papoulis and Pillai, **Probability, Random variables and Stochastic processes**, 4th Edition, McGraw-Hill, 2002.